

BASICS OF MECHANICAL ENGINEERING

**Course Code: 25ESC-ME-103H | Category:
Engineering Science Course | Semester I/II**

**L-T-P: 3-0-0 | Credits: 3 | Complete Theoretical
Notes (Units I–IV)**

UNIT I

PART A: INTRODUCTION TO COMMONLY USED MACHINE TOOLS

1.1 Introduction

A **machine tool** is a power-driven device used to shape, cut, or form metal (or other rigid materials) by removing excess material in the form of chips.

Machine tools form the backbone of the manufacturing industry and are broadly classified as

generating type (lathe, milling) and **forming type** (broaching).

1.2 Lathe Machine

Definition: A lathe is a machine tool that removes unwanted material from a rotating workpiece by means of a cutting tool that is fed either parallel or perpendicular to the axis of rotation.

Principle of working: The workpiece is held and rotated between centers (or in a chuck); the tool, mounted on a tool post, is moved linearly to remove material and produce cylindrical, tapered, or contoured surfaces.

Main parts of a lathe:

- **Bed** – heavy base that supports all other parts and carries guideways.
- **Headstock** – contains the spindle, gears, and drive mechanism; holds and rotates the workpiece.
- **Tailstock** – supports the other end of the workpiece; can hold a dead/live center or drill.
- **Carriage** – consists of saddle, cross-slide, compound rest, and tool post; carries and moves the cutting tool.
- **Feed mechanism** – lead screw and feed rod transmit motion from headstock to carriage for

automatic feed/thread cutting.

Lathe specification: Height of centers above bed, length of bed, swing diameter over bed, swing diameter over carriage, and horsepower of motor.

Operations performed on a lathe:

Operation	Description
Turning	Reducing diameter of workpiece by removing material parallel to axis
Facing	Producing a flat surface perpendicular to the axis
Taper turning	Producing a conical surface (by swiveling compound rest, tailstock offset, or taper attachment)
Thread cutting	Cutting helical grooves using lead screw
Knurling	Producing a diamond-shaped rough pattern for gripping
Drilling/Boring	Enlarging or finishing a hole using tailstock-held tool
Parting off	Cutting the finished workpiece from the stock

1.3 Shaper Machine

Definition: A reciprocating machine tool in which the ram, carrying a single-point cutting tool, moves back and forth over a stationary workpiece to produce flat, contoured, or grooved surfaces.

Working principle: Cutting occurs only during the **forward (cutting) stroke**; the **return stroke is idle**, and is made faster than the cutting stroke using a **Quick Return Mechanism (Crank and Slotted Link Mechanism)** to save time.

Main parts: Base, column, cross rail, saddle, table, ram, tool head.

Types of shapers: Standard (horizontal), Vertical shaper, Universal shaper, Draw-cut shaper, Traveling head shaper.

Operations: Machining flat horizontal/vertical/angular surfaces, cutting slots, keyways, grooves, and gear teeth (in small batches).

1.4 Planer Machine

Definition: A machine tool similar to a shaper but designed for very large and heavy workpieces, where the **workpiece (on the table) reciprocates** while the **tool remains stationary** (opposite of a shaper).

Difference between Shaper and Planer:

Shaper	Planer
Tool reciprocates, work is stationary	Work reciprocates, tool is stationary
Used for small/medium workpieces	Used for large, heavy workpieces
Single tool head	Multiple tool heads possible, higher output

Types: Double housing planer, Open side planer, Pit planer, Edge/plate planer.

1.5 Milling Machine

Definition: A machine tool in which material is removed by feeding the workpiece against a rotating multi-point cutter.

Types:

- **Horizontal milling machine** – cutter mounted on a horizontal arbor.
- **Vertical milling machine** – spindle vertical, cutter mounted directly on the spindle nose.
- **Universal milling machine** – table can swivel, capable of helical milling.

Milling cutters: Plain, side, face, end mill, slab, form cutters, slitting saw.

Operations: Plain (slab) milling, face milling, angular milling, form milling, gang milling, gear cutting, slot & keyway milling.

1.6 Drilling Machine

Definition: A machine tool used to originate or enlarge a round hole in a workpiece using a rotating cutting tool called a **drill (twist drill)**.

Types: Portable drilling machine, Sensitive (bench) drilling machine, Upright drilling machine, Radial drilling machine, Gang drilling machine, Multiple spindle drilling machine.

Operations:

- **Drilling** – originating a hole.
- **Reaming** – finishing/sizing an existing hole accurately.
- **Boring** – enlarging an existing hole.
- **Counterboring** – enlarging the end of a hole to a certain depth (for bolt heads).
- **Countersinking** – making a conical enlargement at hole mouth (for screw heads).
- **Tapping** – cutting internal threads.
- **Spot facing** – smoothing a flat seat around a hole.

1.7 Slotter (Slotting Machine)

Definition: A reciprocating machine tool in which the ram holding the tool moves **vertically up and down** to cut internal/external keyways, slots, and irregular internal profiles (e.g., internal gears, splines) that cannot be produced on a shaper.

Working: Cutting occurs on the **downward stroke**; quick return occurs on the upward stroke (via Whitworth quick return mechanism).

1.8 Introduction to Metal Cutting

Definition: Metal cutting is a process of removing unwanted material from a workpiece in the form of chips with the help of a cutting tool to obtain the desired shape, size, and finish.

Tool geometry: Rake angle (positive/negative/zero), clearance angle, cutting edge angle — these govern chip flow, cutting force, and surface finish.

Types of cutting:

- **Orthogonal cutting** – cutting edge perpendicular to the direction of tool travel (2-D cutting, single force analysis).
- **Oblique cutting** – cutting edge inclined at an angle other than 90° to the direction of motion (3-D cutting, more realistic).

Types of chips formed:

1. **Continuous chips** – formed while machining ductile materials at high speed with proper rake angle; gives good surface finish.
2. **Discontinuous chips** – formed while machining brittle materials (cast iron) or ductile materials at low speed.
3. **Continuous chips with built-up edge (BUE)** – formed at low/medium speed with insufficient rake angle in ductile materials; poor finish.

Other concepts: Tool life (time between two successive grindings), tool wear (flank wear, crater wear), cutting fluids (coolant + lubricant, e.g., soluble oil) used to reduce friction and temperature.

PART B: BASIC CONCEPTS OF THERMODYNAMICS

1.9 Introduction

Thermodynamics is the branch of science that deals with the relationships between heat, work, and the properties of systems, and the conversion of energy from one form to another.

1.10 System, Boundary, and Surroundings

- **System:** A specific quantity of matter or a region of space chosen for study.
- **Boundary:** The real or imaginary surface separating the system from surroundings.
- **Surroundings:** Everything external to the system.

Types of systems:

Type	Mass transfer	Energy transfer	Example
Open system	Yes	Yes	Air compressor, turbine
Closed system	No	Yes	Gas in a piston-cylinder
Isolated system	No	No	Thermos flask (ideal)

1.11 Properties, State, Path, Process, Cycle

- **Property:** A characteristic of a system used to describe its state (e.g., pressure, volume, temperature).
 - **Intensive property:** independent of mass of system (pressure, temperature, density).

- **Extensive property:** dependent on mass of system (volume, energy, mass) – becomes intensive when expressed per unit mass (specific volume, specific energy).
- **State:** The condition of a system as identified by its properties at a given instant.
- **Path:** The series of states through which a system passes during a process.
- **Process:** The transformation of a system from one state to another (e.g., isothermal, isobaric, isochoric, adiabatic).
- **Cycle:** A series of processes after which the system returns to its original state (net change in properties = zero).

1.12 Work and Heat

Work (W): Energy transfer between system and surroundings due to a driving force, other than a temperature difference (e.g., mechanical work, electrical work).

- **Sign convention:** Work done **by** the system is taken as **positive**; work done **on** the system is **negative**.
- **Displacement (pdV) work:** $W = \int p \cdot dV$, done when the boundary of a system moves (e.g., piston-cylinder expansion/compression).

Heat (Q): Energy transfer between a system and surroundings due to temperature difference alone.

- **Sign convention:** Heat supplied **to** the system is **positive**; heat rejected **by** the system is **negative**.

Both heat and work are **path functions** (not properties) – their value depends on the path followed, not just end states; hence they are inexact differentials (δQ , δW), unlike properties which are exact differentials (dU , dV).

1.13 Temperature and the Zeroth Law

Temperature is a measure of the degree of hotness or coldness of a body, and determines the direction of heat flow.

Zeroth Law of Thermodynamics: *"If two bodies are each in thermal equilibrium with a third body, then they are in thermal equilibrium with each other."*

- This law forms the basis for temperature measurement (thermometry), since it justifies using a third body (thermometer) to compare the temperature of two other bodies.

1.14 First Law of Thermodynamics

Statement: *"When a system undergoes a cyclic process, the net heat transfer is equal to the net work transfer."* Mathematically: $\oint \delta Q = \oint \delta W$

For a non-cyclic (closed system) process: $\delta Q = dU + \delta W$, i.e., heat supplied to a system is used partly to increase its internal energy and partly to do external work.

Internal Energy (U): The energy stored within a system due to molecular motion (kinetic) and molecular configuration/bonding (potential). It is a **point function/property** — depends only on the state, not the path.

Enthalpy (H): A property defined as $H = U + pV$, representing the total energy of a flow system (internal energy + flow work). Used extensively in analyzing open systems (turbines, compressors, nozzles) where flow work is significant.

- Specific enthalpy: $h = u + pv$

Significance: The First Law is essentially the **law of conservation of energy** applied to thermodynamic systems — energy can neither be created nor destroyed, only converted from one form to another.

1.15 Second Law of Thermodynamics

The First Law does not indicate the **direction** in which a process can occur, nor does it distinguish between heat and work quality. The Second Law addresses this.

Kelvin-Planck Statement: *"It is impossible to construct a heat engine operating in a cycle that produces no effect other than the extraction of heat from a single reservoir and its complete conversion into an equivalent amount of work."* (No engine can have 100% efficiency.)

Clausius Statement: *"It is impossible to construct a device operating in a cycle that produces no effect other than the transfer of heat from a colder body to a hotter body."* (Heat cannot flow spontaneously from cold to hot without external work input, as in a refrigerator.)

Both statements are equivalent — violation of one implies violation of the other.

Entropy (S): A thermodynamic property that measures the **degree of molecular disorder/randomness** of a system, and indicates the direction and feasibility of a process.

- Mathematically, change in entropy: $dS = \delta Q_{\text{rev}} / T$
- For any real (irreversible) process, entropy of an isolated system always **increases**: $\Delta S \geq 0$ (Principle of increase of entropy).
- Entropy is a measure of the unavailability of energy to do useful work.

1.16 Third Law of Thermodynamics

Statement: *"The entropy of a pure, perfect crystalline substance is zero at absolute zero temperature (0 K)."*

This law provides the basis for the absolute measurement of entropy and confirms that absolute zero temperature can be approached but never actually attained.

1.17 Illustrative Problem Approach

Typical numerical problems in this unit involve:

- Applying **$Q = \Delta U + W$** for closed systems undergoing constant pressure/volume/temperature processes.
- Calculating **pdV work** for isothermal ($W = p_1 V_1 \ln(V_2/V_1)$), isobaric ($W = p(V_2 - V_1)$), and polytropic ($pV^n = C$) processes.
- Finding change in enthalpy ($\Delta h = C_p \Delta T$) and internal energy ($\Delta u = C_v \Delta T$) for ideal gases.
- Determining efficiency of a heat engine: **$\eta = W_{\text{net}}/Q_{\text{in}} = 1 - (Q_{\text{reject}}/Q_{\text{supplied}})$** .

UNIT II

PART A: REFRIGERATION AND AIR CONDITIONING

2.1 Introduction

Refrigeration is the process of removing heat from a low-temperature region (refrigerated space) and rejecting it to a high-temperature region (surroundings), thereby maintaining the space at a temperature lower than its surroundings. **Air conditioning** additionally controls temperature, humidity, cleanliness, and air motion for human comfort or industrial processes.

Unit of refrigeration – Ton of Refrigeration (TR): The rate of heat removal required to freeze one (short/long) ton of water at 0°C into ice at 0°C in 24 hours.

$1 \text{ TR} \approx 3.5 \text{ kW} \approx 210 \text{ kJ/min} \approx 12,000 \text{ Btu/hr.}$

2.2 Rating of Refrigeration Machines

Refrigeration machines are rated in terms of their **refrigerating capacity** (in TR or kW), representing the rate at which they can extract heat from the refrigerated space.

2.3 Coefficient of Performance (COP)

COP is the ratio of desired refrigerating effect to the net work input required to produce that effect:

COP(refrigerator) = Refrigerating Effect (Q_L) / Work Input (W)

COP(heat pump) = Heat Rejected (Q_H) / Work Input (W) = COP(refrigerator) + 1

A higher COP indicates a more efficient refrigeration system. Unlike engine efficiency, COP can be greater than 1.

2.4 Simple Vapour Compression Refrigeration Cycle

This is the most widely used refrigeration cycle, consisting of four main components:

Component	Function
Compressor	Compresses low-pressure refrigerant vapour to high pressure and temperature
Condenser	Rejects heat from high-pressure vapour to surroundings; refrigerant condenses to liquid
Expansion valve (throttle valve)	Reduces pressure and temperature of liquid refrigerant (throttling process)

Component	Function
Evaporator	Refrigerant absorbs heat from the refrigerated space and evaporates, producing the cooling effect

Working cycle (four processes):

1. **Compression (1–2):** Reversible adiabatic (isentropic) compression of dry saturated vapour from evaporator pressure to condenser pressure.
2. **Condensation (2–3):** Constant pressure heat rejection in the condenser; vapour condenses to saturated liquid.
3. **Expansion (3–4):** Irreversible throttling (constant enthalpy) through the expansion valve; pressure and temperature drop sharply.
4. **Evaporation (4–1):** Constant pressure heat absorption in the evaporator; liquid evaporates to dry saturated vapour, producing the refrigerating effect.

This cycle is conveniently represented on a **p-h (pressure-enthalpy) diagram** and a **T-s (temperature-entropy) diagram**, which are used to determine refrigerating effect, work of compression, and COP.

2.5 Psychrometry

Psychrometry is the science dealing with the properties of moist air (mixture of dry air and water vapour) and their use in analyzing air conditioning processes.

Key psychrometric properties:

- **Dry Bulb Temperature (DBT):** Temperature of air measured by an ordinary thermometer, unaffected by moisture.
- **Wet Bulb Temperature (WBT):** Temperature recorded by a thermometer whose bulb is covered with a wet wick exposed to air flow; indicates evaporative cooling effect.
- **Dew Point Temperature (DPT):** Temperature at which moisture in the air begins to condense when cooled at constant pressure.
- **Humidity Ratio (Specific Humidity):** Mass of water vapour per unit mass of dry air (kg vapour/kg dry air).
- **Relative Humidity (RH):** Ratio of actual mass of water vapour in air to the mass required for saturation at the same temperature, expressed as a percentage.

Psychrometric chart: A graphical representation plotting DBT (x-axis) against humidity ratio (y-axis), also showing RH, WBT, DPT, specific volume, and enthalpy lines. It is used to quickly determine

unknown psychrometric properties when two are known, and to analyze processes like sensible heating/cooling, humidification, dehumidification, and mixing of air streams.

2.6 Human Comfort

Human comfort in air conditioning depends on:

- **Effective temperature** – an index combining the effects of DBT, humidity, and air motion on the sensation of warmth felt by the human body.
 - Factors affecting comfort: air temperature, relative humidity, air velocity, radiant heat, occupant's clothing and activity level.
 - The **comfort zone** is generally around 22–25°C DBT with 40–60% RH for sedentary occupants.
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PART B: HYDRAULIC TURBINES AND PUMPS

2.7 Introduction

A **hydraulic turbine** is a rotary machine that converts the kinetic and pressure energy of flowing/falling water into mechanical (shaft) energy, which is generally used to drive an electrical generator (hydroelectric power plants). Key terms: **Gross head**

(difference in water level between reservoir and tailrace), **Net/effective head** (gross head minus losses), **Discharge** (volume flow rate of water).

2.8 Classification of Turbines

Basis	Types
Action of water on blades	Impulse turbine (Pelton wheel) and Reaction turbine (Francis, Kaplan)
Direction of flow	Tangential, Radial, Axial, Mixed flow
Head available	High head (Pelton), Medium head (Francis), Low head (Kaplan)
Specific speed	Low, medium, high specific speed turbines

2.9 Pelton Wheel (Impulse Turbine)

Construction: Consists of a rotating wheel (runner) fitted with a series of double-hemispherical (spoon-shaped) **buckets** on its periphery, a **nozzle** that directs a high-velocity jet of water tangentially onto the buckets, and a **spear/needle** to regulate flow through the nozzle.

Working principle: The entire available head is converted into kinetic energy by the nozzle before the jet strikes the buckets — hence pressure remains atmospheric throughout (impulse action). The jet strikes the splitter of the bucket, is deflected through nearly $160\text{--}165^\circ$, and the change in momentum produces the driving force that rotates the wheel. Used for **high head, low discharge** installations.

2.10 Francis Turbine (Reaction Turbine)

Construction: Consists of a **spiral (scroll) casing**, **guide vanes/wicket gates** (fixed to a stay ring, regulate water flow onto the runner), and a **runner** with curved vanes through which water flows radially inward and axially downward (mixed flow) to the draft tube.

Working principle: Water enters the runner under pressure and partly converts pressure energy into kinetic energy within the runner itself (reaction action), unlike the Pelton wheel. The runner is always full of water, unlike the impulse turbine. Used for **medium head, medium discharge** applications.

2.11 Kaplan Turbine (Reaction Turbine)

Construction: An axial-flow reaction turbine with an **adjustable-pitch propeller-type runner** (similar to a

ship's propeller), guide vanes, and a draft tube. Water flows parallel to the shaft axis.

Working principle: Similar reaction principle as Francis turbine, but water flows axially through the runner. The adjustable runner blades allow efficient operation over a wide range of loads. Used for **low head, high discharge** installations.

Comparative summary:

Turbine	Type	Head	Flow direction
Pelton	Impulse	High (>250 m)	Tangential
Francis	Reaction	Medium (60–250 m)	Radial → Axial (mixed)
Kaplan	Reaction	Low (<60 m)	Axial

2.12 Classification of Water Pumps

A **pump** converts mechanical energy into hydraulic (pressure/kinetic) energy to move/lift water.

(a) Centrifugal Pump:

- **Construction:** Impeller (rotating vanes) enclosed in a casing (volute or diffuser type), connected to a suction pipe and delivery pipe.
- **Working:** The rotating impeller imparts centrifugal (kinetic) energy to the water, which is then converted into pressure energy in the casing as the water moves outward and is discharged through the delivery pipe.
- **Classification:** By casing type (volute, diffuser/turbine), number of impellers (single-stage, multi-stage), and direction of flow (radial, axial, mixed).

(b) Reciprocating Pump:

- **Construction:** A piston/plunger moving inside a cylinder, connected via a crank and connecting rod to a driving shaft; fitted with suction and delivery valves (non-return valves).
- **Working:** During the suction stroke, the piston moves backward creating a vacuum that draws water in through the suction valve; during the delivery stroke, the piston moves forward, closing the suction valve and forcing water out through the delivery valve — positive displacement action.
- **Classification:** Single-acting (delivers water once per revolution) and double-acting (delivers water twice per revolution); Single-cylinder and multi-cylinder types.

UNIT III

PART A: POWER TRANSMISSION METHODS AND DEVICES

3.1 Introduction

Power transmission is the process of transferring mechanical power from a driving shaft (e.g., motor/engine) to a driven shaft/machine, generally over some distance, and often with a change in speed or torque. Common methods include belt, rope, chain, and gear drives.

3.2 Belt Drives

Types of belts: Flat belt, V-belt, and circular (round) belt.

Types of belt drives:

- **Open belt drive** – used when both pulleys rotate in the **same direction**, shafts parallel.
- **Cross belt drive** – used when pulleys must rotate in **opposite directions**; belt crosses itself, causing more wear.

Key concepts:

- **Velocity ratio** = Speed of driven pulley / Speed of driving pulley = Diameter of driver / Diameter of driven (ignoring slip and belt thickness).
- **Slip**: Relative motion between belt and pulley surface due to insufficient friction, reducing velocity ratio.
- **Creep**: Slight relative movement of the belt over the pulley due to alternate extension and contraction of the belt as it moves from tight side to slack side (elastic property of belt).

V-belts are preferred over flat belts for shorter center distances and higher power transmission due to wedging action in the pulley groove, providing greater grip with less slip.

3.3 Rope Drives

Used when large amounts of power must be transmitted over long distances.

- **Fibre ropes** (cotton, hemp, manila) – used for moderate power transmission at low speeds.
- **Wire ropes** – used for high power transmission, cranes, elevators, and haulage, where high strength is required.

3.4 Chain Drives

A positive drive using a series of interconnected links engaging with toothed wheels called **sprockets**.

- **Roller chain:** Widely used in industrial machinery, bicycles, and automobiles.
- **Silent (inverted tooth) chain:** Used where smooth, quiet operation at high speed is required.

Advantages of chain drive over belt drive: No slip (positive drive), compact, higher efficiency, capable of transmitting more power, works well at both low and high speeds without loss of efficiency.

3.5 Gear Drives

Used for positive transmission of power/motion between shafts with an exact velocity ratio, over short center distances.

Types of gears:

- **Spur gears** – teeth parallel to the axis; used for parallel shafts.
- **Helical gears** – teeth inclined at an angle (helix angle); smoother, quieter operation; used for parallel or crossed shafts.
- **Bevel gears** – conical shape; used to transmit power between intersecting shafts (usually at 90°).
- **Worm gears** – worm (screw) meshing with a worm wheel; used for large speed reduction

between non-intersecting, perpendicular shafts.

3.6 Clutches

A **clutch** is a mechanical device used to connect and disconnect the driving shaft from the driven shaft at the will of the operator, without stopping the driving shaft — essential for smooth engagement/disengagement of power (e.g., in automobiles).

Types and working:

- **Single-plate friction clutch:** A single friction plate (lined with friction material) is sandwiched between the flywheel and a pressure plate; spring pressure holds them together, transmitting torque by friction. Depressing the clutch pedal disengages the plate.
- **Multi-plate friction clutch:** Uses multiple friction and pressure plates alternately, increasing the friction surface area to transmit higher torque within a compact size; commonly used in motorcycles and heavy vehicles.
- **Cone clutch:** Uses a cone-shaped friction surface (male cone on driving shaft engaging a female cone on driven shaft) — provides more torque capacity than a flat plate for a given axial force, due to wedging action, but is more prone to wear and drag.

- **Centrifugal clutch:** Engages automatically as rotational speed increases — spring-loaded friction shoes fly outward due to centrifugal force at higher speeds and grip the driven drum, engaging the clutch without any pedal/lever operation (common in mopeds, go-karts).
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PART B: STRESSES AND STRAINS

3.7 Introduction

When an external force (load) is applied to a solid body, internal resisting forces are developed within the material, and the body undergoes deformation. The study of stress and strain forms the basis of the **Strength of Materials**.

3.8 Stress

Stress is defined as the internal resisting force developed per unit cross-sectional area of a material when subjected to an external load.

Stress (σ or τ) = Force (P) / Cross-sectional Area (A),
units: N/m² (Pascal) or N/mm² (MPa).

Types of stress:

- **Tensile stress:** Develops when a body is subjected to two equal and opposite pulling

forces, causing elongation.

- **Compressive stress:** Develops when a body is subjected to two equal and opposite pushing forces, causing shortening.
- **Shear stress:** Develops when equal, opposite, parallel forces act tangentially on a section, causing one layer to slide over an adjacent layer.

3.9 Strain

Strain is the ratio of change in dimension to the original dimension of a body when subjected to a load – a dimensionless quantity.

Types of strain:

- **Linear (longitudinal) strain:** Change in length / Original length ($\epsilon = \delta L/L$).
- **Lateral strain:** Change in lateral (transverse) dimension / Original lateral dimension – occurs perpendicular to the applied load direction.
- **Volumetric strain:** Change in volume / Original volume.
- **Shear strain:** Angular deformation (change of angle, in radians) produced by shear stress.

3.10 Poisson's Ratio

When a body is stretched, it elongates in the direction of load (longitudinal strain) but simultaneously

contracts in the perpendicular directions (lateral strain). The ratio of lateral strain to longitudinal strain (within elastic limit) is a constant for a given material, called **Poisson's ratio (μ or ν)**:

$$\mu = \text{Lateral strain} / \text{Longitudinal strain}$$

For most metals, μ lies between 0.25 and 0.35.

3.11 Stress-Strain Diagram (for Mild Steel – Ductile Material)

When a ductile specimen (e.g., mild steel) is gradually loaded in tension until fracture, and stress is plotted against strain, the following characteristic points are observed:

- **Proportional limit (A):** The point up to which stress is directly proportional to strain (Hooke's law is valid).
- **Elastic limit (B):** The maximum stress up to which the material returns to its original shape upon removal of load (no permanent deformation).
- **Upper & Lower Yield Point (C, D):** The point at which the material begins to deform plastically with little or no increase in load; a sudden drop/fluctuation in stress is observed.
- **Ultimate stress (E):** The maximum stress the material can withstand – corresponds to the peak

of the curve; **necking** (local reduction in cross-section) begins here.

- **Breaking/Fracture stress (F):** The stress at which the specimen finally fractures – apparent stress is lower than ultimate stress because it is calculated on the original area, while the actual (true) cross-section has reduced significantly due to necking.

(Brittle materials like cast iron do not exhibit a distinct yield point and fracture soon after the elastic limit, with very little plastic deformation.)

3.12 Hooke's Law

Statement: "Within the elastic limit, stress is directly proportional to strain."

$\sigma \propto \epsilon \Rightarrow \sigma = E \cdot \epsilon$, where **E** is a constant of proportionality called the **Modulus of Elasticity** (**Young's Modulus**).

3.13 Elastic Constants

- **Young's Modulus (E):** Ratio of tensile/compressive stress to the corresponding linear strain – a measure of a material's stiffness in tension/compression.
- **Modulus of Rigidity / Shear Modulus (G or C):** Ratio of shear stress to shear strain – measures resistance to shear deformation.

- **Bulk Modulus (K):** Ratio of volumetric (hydrostatic) stress to volumetric strain – measures resistance to uniform compression.

Relationship between elastic constants:

- $E = 2G(1 + \mu)$
- $E = 3K(1 - 2\mu)$
- $E = 9KG / (3K + G)$

3.14 Mechanical Properties of Metals

- **Elasticity:** Ability of a material to return to its original shape after removal of load.
- **Plasticity:** Ability of a material to undergo permanent deformation without fracture.
- **Ductility:** Ability of a material to be drawn into thin wires (large plastic deformation under tension), e.g., copper, mild steel.
- **Malleability:** Ability of a material to be hammered/rolled into thin sheets (large plastic deformation under compression), e.g., gold, aluminium.
- **Toughness:** Ability of a material to absorb energy and withstand shock/impact loads without fracturing.
- **Hardness:** Resistance of a material to indentation, scratching, abrasion, or wear.

- **Brittleness:** Property of a material to fracture with little or no plastic deformation (opposite of ductility), e.g., cast iron, glass.
 - **Stiffness:** Resistance of a material to elastic deformation (related to modulus of elasticity).
 - **Resilience:** Capacity of a material to absorb energy elastically and release it upon unloading.
 - **Fatigue strength:** Ability of a material to withstand repeated/cyclic loading without failure.
 - **Creep:** Slow, progressive plastic deformation of a material under a constant (sustained) load, usually at elevated temperature, over a long period of time.
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UNIT IV

INTRODUCTION TO MANUFACTURING SYSTEMS AND NUMERICAL CONTROL

4.1 Introduction to Manufacturing Systems

A **manufacturing system** is an integrated combination of processes, machine tools, material handling, and personnel designed to perform one or more

processing/assembly operations to convert raw materials into finished products.

Types of manufacturing systems (based on production volume/variety):

- **Job shop production:** Small quantities of a wide variety of custom products; general-purpose machines; highly skilled labour.
- **Batch production:** Limited quantities of a product manufactured periodically in batches; moderately flexible machines.
- **Mass production:** Large quantities of a standardized product manufactured continuously; special-purpose machines, dedicated production lines.
- **Continuous/Flow production:** Very high volume, standardized products produced in an uninterrupted flow (e.g., chemical/process industries).

4.2 Fundamentals of Numerical Control (NC)

Definition: Numerical Control (NC) is a method of automatically controlling the movements and operations of a machine tool using a series of coded numerical/alphabetic instructions (a "part program"), rather than manual operation by a machinist.

Basic components of an NC system:

1. **Part program:** A set of coded instructions (in a standard code, e.g., G-codes/M-codes) representing the machining sequence and tool paths, prepared according to the geometry of the part and the machining operations required.
2. **Machine Control Unit (MCU):** The "brain" of the NC system; reads, interprets, and converts the coded part program into electrical signals to control machine tool actuators (motors, drives).
3. **Machine tool:** The actual machine (lathe, milling machine, etc.) equipped with servo motors/drives that execute the motions commanded by the MCU.

4.3 Advantages of NC Systems

- High accuracy and repeatability of machined parts.
- Reduced dependency on operator skill; reduced human error.
- Increased productivity through reduced setup and machining time.
- Flexibility to manufacture complex shapes and quickly switch between different jobs by simply changing the part program.
- Reduced scrap and rework, better quality control.
- Reduced lead time for producing new components.

4.4 Classification of NC Systems

(a) Based on motion control (type of path):

- **Point-to-Point (PTP) Control:** The tool moves from one point to another without controlling the path taken between the points; only the end position accuracy matters. Used for drilling, spot welding, punching operations.
- **Straight Cut (Line) Control:** The tool can move along axes at a controlled feed rate in straight lines (parallel to machine axes), performing cutting along the way, but cannot follow angular/curved paths precisely.
- **Contouring (Continuous Path) Control:** The tool path is continuously controlled along a curved or angular contour by simultaneous coordinated control of two or more axes; used for milling complex profiles, cams, dies.

(b) Based on control loop:

- **Open loop system:** No feedback mechanism to verify actual tool position against the commanded position (uses stepper motors); simpler and cheaper but less accurate.
- **Closed loop system:** Uses feedback devices (position/velocity transducers, encoders) to continuously compare actual tool position with

the commanded position and correct any error;
more accurate and reliable.

4.5 Comparison of NC and CNC

Aspect	NC (Numerical Control)	CNC (Computer Numerical Control)
Control	Hard-wired logic circuits (fixed)	Dedicated computer with software (flexible)
Program storage	Punched tape/cards read repeatedly for each job (no on-board memory)	Program stored in computer memory; can be edited/reused instantly
Flexibility	Limited; modifying program requires new tape	High; programs can be edited, stored, and modified at the machine
Editing	Not possible at the machine;	Possible directly at the

Aspect	NC (Numerical Control)	CNC (Computer Numerical Control)
	requires external re-punching	machine controller
Diagnostics	Not available	Built-in self-diagnostic capability
Accuracy over time	Tape wear can cause errors	Higher and more consistent accuracy, no tape wear
Cost	Relatively lower initial cost, higher operating cost	Higher initial cost, but more efficient in the long run
Multiple part programs	Cannot store multiple programs simultaneously	Can store and manage several part programs in memory

QUICK REVISION SUMMARY

Unit	Key Themes
I	Machine tools (lathe, shaper, planer, milling, drilling, slotter, metal cutting) + Laws of Thermodynamics (Zeroth, First, Second, Third), internal energy, enthalpy, entropy
II	Refrigeration (COP, vapour compression cycle, psychrometry, comfort) + Hydraulic turbines (Pelton, Francis, Kaplan) and pumps (centrifugal, reciprocating)
III	Power transmission (belt, rope, chain, gear, clutch) + Stress-strain fundamentals (Hooke's law, Poisson's ratio, elastic constants, mechanical properties)
IV	Manufacturing systems, NC fundamentals, classification (PTP/straight-cut/contouring, open/closed loop), NC vs CNC

Reference Text/Books for further reading: Elements of Mechanical Engineering (R.K. Rajput; D.S. Kumar), Engineering Thermodynamics (P.K. Nag), Refrigeration & Airconditioning (Arora & Domkundwar), Strength of Materials (Popov; G.H. Ryder), Hydraulic Machines (Jagdish Lal).